Simple Video Editor

Product + Technical Specification (CapCut-style, simplified)

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Purpose: hand-off document for an AI coding agent (Codex) to build the application end to end.

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Kit contents

- 01_VIDEO_EDITOR_SPEC.pdf - this document (what to build and how)
- 02_PROMPT_FOR_CODEX.md - the main prompt to paste into Codex
- 03_AGENTS.md - rules file; put it in the project root as AGENTS.md
- 04_MILESTONES.md - step-by-step build plan with a prompt for each stage
- 05_project.schema.json - JSON schema of the project file (.vedit)

1. Product Overview

A desktop video editor that feels like a lighter CapCut. The user creates a project, imports videos, images and audio, arranges them on a multi-track timeline, adds styled text, previews in real time, and exports a finished video (up to 4K at 60 fps), an audio-only file, or both.

1.1 Goals

- Multi-track timeline: several video/image tracks (layers) and several audio tracks.
- Text layers with a full styles panel.
- Project creation, a start screen with recent projects, autosave.
- Export: video + audio combined, audio only, up to 3840x2160 at 60 fps.
- Simple, clean, dark UI. Fast to learn.

1.2 Non-goals for v1 (do NOT build yet)

- Keyframe animation graphs, motion tracking, green screen, AI features, auto-captions.
- Cloud sync, accounts, collaboration, mobile app.
- Plugin system, third-party effect packs, LUT import.
- Complex transitions (only a basic cross-fade and fade in/out in v1).

1.3 Your six requirements mapped to this spec

#	Requirement	Where covered
1	Video and images on multiple lines and layers	Sec 4 (timeline), 6 (data model), 7 (preview)
2	Multiple audio layers	Sec 4.3, 8
3	Text editing, styles panel	Sec 5.4, 9
4	Export video+audio combined, and separate audio	Sec 10
5	Export 4K at 60 fps	Sec 10.2, 10.3
6	Project creation and start interface	Sec 5.1, 5.2, 11

2. Recommended Tech Stack

The editor is a **desktop app** (Electron) because 4K/60 export needs reliable native FFmpeg, local file access and hardware encoders. A pure browser app can export 4K but is far less reliable with long or heavy projects. Since you are a web developer, the UI is plain React and TypeScript.

Layer	Choice	Why
Shell	Electron (latest stable) + electron-builder	Native file access, bundles FFmpeg, Windows/Mac/Linux
UI	React 18 + TypeScript + Vite	Familiar, fast dev
Styling	Tailwind CSS + Radix UI primitives	Quick, accessible dark UI
State	Zustand + Immer, with undo/redo history	Simple, immutable edits

Layer	Choice	Why
Preview	HTML5 video/audio elements synced to a master clock, composited on a Canvas 2D (upgrade to PixiJS/WebGL if slow)	Real-time layered preview
Timeline UI	Custom React component, HTML/CSS (virtualised for long projects)	Full control of drag/trim/snap
Waveforms	Web Audio API decode, cached to PNG/array per asset	Audio track display
Media processing	FFmpeg + FFprobe binaries (ffmpeg-static / ffprobe-static)	Export, thumbnails, proxies, probing
Text render	Canvas 2D in renderer, exported as transparent PNG for FFmpeg overlay	WYSIWYG: text looks identical in preview and export
Storage	Project JSON file (.vedit) + project folder with cache	Portable, human-readable
Testing	Vitest (logic), Playwright (UI smoke), FFmpeg export golden tests	Catch regressions

Fonts: bundle 8-10 free Google fonts (Inter, Poppins, Montserrat, Roboto, Oswald, Bebas Neue, Playfair Display, Lobster, Pacifico, Roboto Mono) and also list system fonts.

3. Architecture

```
Electron Main process (Node)
  - window + menus, file dialogs, recent projects
  - ffmpeg service: probe, thumbnails, waveform, proxy, EXPORT
  - project file IO (save, autosave, load, migrate)
        ^  IPC (typed, contextBridge, no nodeIntegration)  v
Renderer (React)
  - Store (project state, selection, playhead, undo stack)
  - Timeline UI | Preview canvas | Media bin | Inspector panels
  - Playback engine (master clock, media element sync)
  - Text rasteriser (canvas -> PNG for export)
```

- Security: contextIsolation ON, nodeIntegration OFF, a small typed preload API only.
- Never block the UI: all FFmpeg work runs in the main process or a child process, with progress events.
- Project state is the single source of truth. Preview and export BOTH read the same project JSON.

4. Timeline Specification (core of the app)

4.1 Tracks

- **Video tracks** (V1, V2, V3 ...): hold video clips, image clips and text clips. Higher track number draws on top. Unlimited tracks (UI default: start with V1 and V2, add via + button).
- **Audio tracks** (A1, A2, A3 ...): hold audio clips. Video clips have their own embedded audio, which is shown as a linked audio part and can be muted or detached.
- Each track: name, mute, solo (audio), hide (video), lock, height. Reorder tracks by drag.
- Text and sticker-like items live on video tracks as clips of type 'text' (so they layer like anything else).

4.2 Clips

Clip type	Properties
video	assetId, start (timeline sec), duration, sourceIn, sourceOut, speed (0.25-4x), volume, mute, position x/y, scale, rotation, opacity, crop (optional), fade in/out
image	assetId, start, duration (default 5s), position, scale, rotation, opacity, fade in/out
text	text content, full style (see 5.4), start, duration, position, scale, rotation, opacity, fade in/out
audio	assetId, start, duration, sourceIn, sourceOut, volume (0-200%), fade in/out, mute

4.3 Timeline behaviours (all required)

- Drag clips along a track and between tracks of the same family (video to video, audio to audio).
- Trim by dragging left/right edges (changes sourceIn/sourceOut for media, duration for image/text).
- Split at playhead (S key / button). Delete (Del), ripple delete (Shift+Del).
- Snapping to playhead, clip edges, markers, and 0 (toggle with magnet button).
- Multi-select (Shift/Ctrl click, drag-box). Move group together.
- Zoom in/out (slider, Ctrl+scroll), horizontal scroll, 'fit to project' button.
- Drag assets from the media bin onto the timeline; dropping on empty space below creates a new track.
- Overlap rule on one track: clips may not overlap on the SAME track (block or push). Overlap across tracks is normal.
- Playhead scrubbing by clicking the ruler or dragging; preview updates live.
- Waveform shown inside audio clips; thumbnails strip inside video clips.
- Undo/redo for every edit (min 100 steps).

5. Screens and UI

5.1 Start screen (Home)

- Left: logo, 'New Project' big button, 'Open Project' button.
- Main: grid of recent projects (thumbnail, name, last edited, duration, resolution). Right-click or menu: rename, duplicate, delete, reveal in folder.
- Search box for projects. Empty state with friendly message.
- Settings: default save folder, theme (dark default), autosave interval.

5.2 New Project dialog

Field	Options / default
Project name	Text, default 'Untitled Project'
Aspect ratio	16:9 (default), 9:16, 1:1, 4:5, 4:3, 21:9, custom
Resolution	720p, 1080p (default), 1440p, 4K (3840x2160 for 16:9; correct equivalents for other ratios)
Frame rate	24, 25, 30 (default), 50, 60
Background color	Color picker, default black

Field	Options / default
Save location	Folder picker, default Documents/VideoEditorProjects

5.3 Editor layout

```
+-------------------------------------------------------------------+
| Top bar: Home | Project name | Undo Redo | Ratio | [Export]       |
+----------------+--------------------------+---------------------+
| Left panel     | Preview canvas           | Right panel         |
| Tabs:          |  (with on-canvas move /  |  Inspector (tabs):  |
|  Media         |   scale / rotate handles) |   Video / Text /    |
|  Audio         |                          |   Audio / Speed     |
|  Text          |  Transport: play, +-1    |                     |
|  (Stickers: v2)|  frame, time, loop       |                     |
+----------------+--------------------------+---------------------+
| Timeline toolbar: split, delete, snap, zoom slider, add track     |
| Ruler + playhead                                                  |
| V3 [ clips ... ]                                                  |
| V2 [ clips ... ]                                                  |
| V1 [ clips ... ]                                                  |
| A1 [ waveform clips ... ]                                         |
| A2 [ waveform clips ... ]                                         |
+-------------------------------------------------------------------+
```

- Resizable panels (drag dividers). Dark theme, accent colour indigo, rounded 8px corners.
- On-canvas handles: drag to move, corner to scale, top handle to rotate, snapping guides to centre/edges.
- Media bin: import button, drag-drop of files from the OS, thumbnails, duration badge, filter by type.

5.4 Text editing and Styles panel

Add text from the Text tab (presets: Title, Subtitle, Caption, Lower-third) or the + Text button. Double-click text on canvas to edit inline. Right panel shows the Text inspector:

Group	Controls
Content	Multiline text box, alignment (left/centre/right), line height, letter spacing
Font	Family dropdown with search, size (px), bold, italic, underline, UPPERCASE toggle
Fill	Solid colour, or 2-stop linear gradient with angle; opacity
Stroke / outline	Colour, width
Shadow	Colour, blur, offset x/y, opacity
Background box	Colour, opacity, padding, corner radius (on/off)
Glow	Colour, blur (on/off)
Transform	Position x/y, scale, rotation, opacity
Animation (basic)	In: none, fade, slide up, pop. Out: none, fade, slide down. Duration slider
Presets	Save current style as preset; 10+ built-in style presets (clicking applies at once)

6. Data Model (project file)

Saved as **MyProject.vedit** (JSON) inside the project folder. A full JSON Schema is in 05_project.schema.json. All times are in seconds (float). Version field is required for future migrations.

```
{
```

```
  "version": 1,
  "id": "uuid", "name": "My Project",
  "settings": { "width": 1920, "height": 1080, "fps": 30,
                "background": "#000000", "sampleRate": 48000 },
  "assets": [
    { "id": "a1", "type": "video|image|audio", "path": "C:/.../clip.mp4",
      "duration": 12.5, "width": 3840, "height": 2160, "fps": 59.94,
      "hasAudio": true, "proxyPath": null, "thumbPath": "cache/a1.jpg" } ],
  "tracks": [
    { "id": "t1", "kind": "video", "name": "V1", "hidden": false,
      "locked": false, "muted": false, "order": 0, "clips": [
      { "id": "c1", "type": "video", "assetId": "a1",
        "start": 0, "duration": 5, "sourceIn": 2, "sourceOut": 7,
        "speed": 1, "volume": 1,
        "transform": { "x": 0.5, "y": 0.5, "scale": 1, "rotation": 0, "opacity": 1 },
        "fadeIn": 0, "fadeOut": 0 } ] },
    { "id": "t2", "kind": "audio", "name": "A1", "muted": false,
      "solo": false, "clips": [ ... ] } ],
  "markers": [ { "time": 4.2, "label": "" } ]
}
```

- Transform x/y are normalised 0-1 relative to the canvas (so changing resolution does not break layouts).
- Asset paths are absolute; if a file is missing on open, show a 'Relink media' dialog.
- Autosave every 30 s to MyProject.autosave.vedit; crash-recovery prompt on next launch.

7. Preview and Playback Engine

- A single master clock (requestAnimationFrame + performance.now) drives the playhead. Media elements follow the clock and are re-synced when drift exceeds 80 ms.
- Each frame: clear canvas to background, then for each video track from bottom to top, draw the active clip (video frame or image) with its transform, opacity and fades; then draw text clips.
- Audio: use Web Audio API (one GainNode per clip, one master bus) so volume, fades, mute and solo mix correctly across multiple audio layers.
- Preview quality selector: Full, Half, Quarter. Auto-use generated **proxy files** (e.g. 960x540 H.264) for 4K sources so scrubbing stays smooth. Export ALWAYS uses the original files.
- Show a rendering/loading state when a clip seeks. Never crash on missing or corrupt media.

8. Audio

- Any number of audio tracks. Supported import: mp3, wav, aac/m4a, ogg, flac, plus audio from video files.
- Per clip: volume 0-200%, fade in/out, mute, split, trim. Per track: mute, solo, volume slider.
- Waveform drawn per clip, cached on disk after first generation.
- Detach audio: right-click a video clip, Detach audio, creating a separate audio clip on an audio track (video clip then muted).
- Master meter in the transport bar (simple peak meter). Master volume slider.
- Mix rule: sum of all audible audio, soft-limited at 0 dBFS to avoid clipping (use FFmpeg alimiter on export).

9. Import and Media Handling

- Video: mp4, mov, mkv, webm, avi. Images: jpg, png, webp, gif (static in v1). Audio as in section 8.
- On import, run ffprobe to read duration, size, fps, audio presence, rotation metadata. Generate a thumbnail and, for sources larger than 1080p, a proxy in the background with a progress badge.
- Variable frame-rate and rotated phone videos must display correctly.
- Imported images have default 5 s duration and are fitted (contain) in the canvas; video fitted by default, with a 'Fill' option.

10. Export (most important technical part)

10.1 Export dialog

Setting	Options
Mode	Video + Audio (default) | Audio only | Both (creates the two files together)
Video format	MP4 (H.264, default) | MP4 (H.265/HEVC) | MOV (ProRes optional, v2) | WebM (VP9)
Resolution	480p, 720p, 1080p, 1440p, 2K, 4K (3840x2160). Export size may differ from the project size; content is scaled to fit
Frame rate	24, 25, 30, 50, 60
Quality / bitrate	Low, Medium, High (default), Max; or custom Mbps. Suggested: 1080p30 = 10 Mbps, 4K30 = 35 Mbps, 4K60 = 60 Mbps
Audio format (audio only)	MP3 (192/320 kbps), WAV (PCM 16-bit), AAC/M4A
Audio in video	AAC 192 kbps, 48 kHz stereo
Range	Whole project | Between in/out markers
Hardware acceleration	Auto-detect: NVENC (NVIDIA), QSV (Intel), AMF (AMD), VideoToolbox (Mac); fall back to libx264/libx265 software
Output	File name + folder, 'Open folder when done' checkbox

10.2 Export pipeline

Export is performed by FFmpeg in the main process from the project JSON. The recommended approach:

- **Step 1 - Build the timeline graph.** Convert tracks and clips into an FFmpeg `filter_complex` script (write it to a temp file and pass with `-filter_complex_script` to avoid command-length limits).
- **Step 2 - Video layers.** Create a base colour source (`color=c=BG:s=WxH:r=FPS:d=TOTAL`). For each video/image clip: trim, setpts (speed), scale with aspect fit, rotate, format=rgba with opacity and fade (alpha), then `overlay` with `enable='between(t,start,end)'` and x/y from the transform. Overlay in track order (V1 first, higher tracks later).
- **Step 3 - Text.** For each text clip, the renderer rasterises the styled text with Canvas 2D to a transparent PNG at export resolution (handles font, stroke, shadow, glow, gradient, background box exactly as in preview). FFmpeg overlays the PNG with fades/slide animation via expressions.
- **Step 4 - Audio.** For each audio clip and each video clip's embedded audio: atrim, asetpts, atempo (speed), volume, afade, adelay (to timeline start). Combine all with `amix=inputs=N:normalize=0`, then apply track and master gain and `alimiter`. Respect mute/solo.

- **Step 5 - Encode.** Video: libx264 / h264_nvenc etc. with `-pix_fmt yuv420p -r FPS -movflags +faststart`. Audio: aac. For audio-only mode skip all video filters and write MP3/WAV/M4A.
- **Step 6 - Progress.** Parse FFmpeg `-progress pipe:1` output (out_time_ms) divided by total duration; send percentage, elapsed and ETA to the UI through IPC. Provide a Cancel button that kills the process and deletes the partial file.
- **Step 7 - Verify.** After export, ffprobe the output: check duration (tolerance 0.1 s), resolution, fps and audio stream; show success or a readable error with an 'Open log' link.

10.3 4K at 60 fps requirements

- Output size 3840x2160 and frame rate 60 must work for any project. Check source frame rates: if the source is 30 fps, FFmpeg duplicates frames via the fps filter (no interpolation in v1).
- Use hardware encoders when available (4K60 software x264 is slow; show an ETA and a note).
- Memory safety: process in one FFmpeg run with streaming (no frame dump to disk). Use `-threads 0`. Warn when free disk space is lower than estimated output size (bitrate x duration).
- Level/profile: H.264 High@5.2 for 4K60, or recommend HEVC for smaller files. Show a warning that some players cannot decode H.264 4K60.
- Export presets (one click): YouTube 1080p30, YouTube 4K60, Instagram Reel (1080x1920 30), Audio MP3 320.

10.4 Separate audio export

- 'Audio only' mode renders the same audio mix without video to MP3, WAV or M4A.
- 'Both' mode produces `name.mp4` and `name.mp3` from one dialog, with one progress bar.
- Optional (nice to have): 'Export each audio track as separate file' (stems).

11. Project Management

- New project dialog (5.2) creates a folder: `Name/Name.vedit`, `Name/cache/` (thumbs, proxies, waveforms), `Name/exports/`.
- Recent projects list stored in app settings (max 30). Project thumbnail updates on save.
- Save (Ctrl+S), Save As, autosave every 30 s, unsaved-changes prompt on close.
- Project settings can be edited later (ratio, resolution, fps, background); clips keep normalised positions.
- Duplicate, rename and delete project from Home.

12. Keyboard Shortcuts

Key	Action
Space	Play / pause
S	Split at playhead
Delete / Backspace	Delete selected
Shift+Delete	Ripple delete

Key	Action
Ctrl+Z / Ctrl+Shift+Z	Undo / redo
Ctrl+C / V / D	Copy / paste / duplicate
Left / Right	Step one frame (Shift = 1 second)
Home / End	Go to start / end
+ / -	Timeline zoom
M	Add marker
N	Toggle snapping
Ctrl+S	Save
Ctrl+E	Open Export
T	Add text

13. Non-functional Requirements

- Preview plays at 30 fps with 3 video layers on a mid-range laptop at Half quality.
- App opens in under 3 seconds; project with 50 clips opens in under 2 seconds.
- No UI freezes: heavy work off the UI thread. Timeline interactions at 60 fps.
- Crash safety: autosave + recovery. Errors shown in plain language, with a log file in the app data folder.
- Platform: Windows 10/11 first-class; macOS and Linux builds should work (electron-builder).
- Code quality: strict TypeScript, ESLint + Prettier, unit tests for timeline logic and the FFmpeg graph builder.

14. Suggested Folder Structure

```
video-editor/
  AGENTS.md
  package.json, vite.config.ts, tsconfig.json, electron-builder.yml
  electron/
    main.ts, preload.ts
    services/ ffmpeg.ts, probe.ts, export.ts, projects.ts, settings.ts
  src/
    app/        routes: Home, Editor
    store/      projectStore.ts, uiStore.ts, history.ts
    timeline/   Timeline.tsx, Track.tsx, Clip.tsx, Ruler.tsx, snapping.ts
    preview/    PreviewCanvas.tsx, playback.ts, compositor.ts, audioEngine.ts
    panels/     MediaBin.tsx, TextPanel.tsx, Inspector/*.tsx
    export/     ExportDialog.tsx, buildFilterGraph.ts, textRasterizer.ts
    lib/        time.ts, geometry.ts, ids.ts, schema.ts
    types/      project.ts
  tests/        unit/ e2e/ golden/
```

15. Build Order (milestones)

M	Deliverable	Done when
M1	Electron+React+TS skeleton, Home screen, New Project dialog, save/load .vedit	Can create, close and reopen a project

M	Deliverable	Done when
M2	Media import (ffprobe, thumbnails), media bin, data model + store + undo	Imported files appear in bin; undo works
M3	Timeline UI: tracks, drag, trim, split, snap, zoom, multi-track	All 4.3 behaviours pass
M4	Preview engine: layered video/image compositing, playback sync, on-canvas handles	Smooth 3-layer playback
M5	Audio: multi-track mixing, waveform, volume, fades, mute/solo, detach	Audio mix correct in preview
M6	Text: add/edit, full styles panel, presets, animation	Text editable with all styles
M7	Export: filter graph builder, FFmpeg run, progress, cancel, audio-only, both	1080p30 export matches preview
M8	4K60 export, hardware encoders, presets, disk/size warnings, proxies	4K60 file verified with ffprobe
M9	Polish: shortcuts, autosave/recovery, relink media, installer build, tests	Installer runs on clean Windows PC

16. Acceptance Tests

- Create a 16:9 4K 60 fps project; import 2 videos, 2 images, 2 audio files, add 3 text clips; arrange on 3 video tracks and 2 audio tracks.
- Play the project: layers stack in correct order; both audio tracks heard together; text appears at the right time.
- Export 4K 60 fps MP4: ffprobe reports 3840x2160, 60 fps, AAC audio; duration equals timeline length (+-0.1 s); a frame at a chosen time visually matches the preview.
- Export audio only (MP3): duration equals timeline length; mute/solo respected.
- Close and reopen the project: everything restores; deleting a source file prompts relink instead of crashing.
- Undo/redo 20 steps across move, trim, split, text edit without corruption.
- Cancel export mid-way: process stops, partial file removed, UI returns to editor.

17. Risks and Notes for the Builder

- Text styling parity between preview and export is the most common bug: always rasterise text with the SAME function for both, never use FFmpeg drawtext for styled text.
- A very long filter_complex can be slow; for projects with many clips consider rendering in segments and concatenating (optimisation after M8).
- Use the original files in export and proxies only in preview.
- Respect codec licensing: bundle a standard FFmpeg build; do not include non-free codecs in redistribution (use native hardware encoders and libx264 where licence allows).
- Keep every milestone shippable: run the app and the tests at the end of each milestone before continuing.